This Tiny Snake Escapes Danger by Rolling Like a Wheel

From: Popular Science Mined: 2026-10-20 5 min read

Snakes have many ways to escape danger. Some bite, some rattle their tails, and some even spit **venom**. But the dwarf reed snake of Southeast Asia has an especially surprising trick: when it feels threatened, it rolls away like a speeding wheel.

An adult dwarf reed snake is usually less than 12 inches long. Yet by rolling, it can move as fast as one foot per second. That is about four times faster than it travels by **slithering**. And it is not a wild, **flailing** escape. The snake can roll along a straight path while keeping its head steady and facing forward.

Scientists at Toho University in Japan wanted to know how it does this. They filmed the snakes with **high-speed cameras**, which can show fast movements in slow motion. They shared what they learned in a study in the journal Current Biology.

The team found that the snake does not spin its whole body in one simple motion. Instead, it whips one side of its body forward toward the ground while a curved part swings down near its head. The researchers call each of these movements a "**stroke**." The snake then switches between its left and right sides with each **stroke**.

It is a bit like twirling a rope in a circle while holding one end in your hand. In this case, the end being held still is the snake's head. This lets the snake keep its eyes and other **senses** upright and pointed ahead, even while it moves quickly.

The study also shows that it is hard to guess what an animal can do just by looking at its body. A snake may be built to slither, but it can still get creative.

Word Treasure

venom -- A poison that some animals, like certain snakes, use to defend themselves or catch food.

slithering -- Moving smoothly along the ground by sliding and twisting the body, the way most snakes travel.

flailing -- Waving or thrashing around wildly without control.

high-speed cameras -- Cameras that take many pictures each second so fast motions can be watched in slow motion.

stroke -- Here it means one repeated swinging movement of the snake's body.

senses -- The body's ways of noticing the world, such as sight, smell and touch.

Background you need

Most snakes move by slithering, pushing the sides of their long bodies against the ground to slide forward.

Small animals that many predators hunt often develop special escape tricks, such as playing dead, jumping or rolling away.

A snake's head holds its eyes and other sense organs, so keeping the head steady helps it see where it is going.

High-speed cameras let scientists slow down very quick movements so they can study exactly how an animal's body moves.

Break it down

LEAD: The dwarf reed snake of Southeast Asia escapes danger by rolling like a wheel
KEY EVENT: Toho University researchers explained how it rolls, in the journal Current Biology
+- WHO: Scientists at Toho University in Japan
+- HOW: Alternating left and right body 'strokes' while the head stays steady
EVIDENCE: High-speed video; rolling reaches one foot per second, about four times its slithering speed
COMPARISON: Like twirling a rope while holding one end still
OPEN QUESTION: What other surprising moves might animals make that their bodies don't seem built for?

Why it matters

The study shows that animals can do things we would never guess from their body shape, which reminds scientists to observe carefully instead of assuming.

Test what you remember

1. What does the dwarf reed snake do when it feels threatened?

- (A) It spits venom
- (B) It digs a hole
- (C) It climbs a tree
- (D) It rolls away like a wheel

2. How long is an adult dwarf reed snake usually?

- (A) Exactly one inch
- (B) Less than 12 inches
- (C) About six feet
- (D) About 20 feet

3. How much faster is rolling than slithering for this snake?

- (A) Twice as fast
- (B) The same speed
- (C) About four times faster
- (D) Ten times slower

4. How did the scientists study the snake's movement?

- (A) With X-ray machines
- (B) By reading old books
- (C) With high-speed cameras
- (D) By measuring its weight

5. What part of its body does the snake keep steady while rolling?

- (A) Its tail
- (B) Nothing stays steady
- (C) Its head
- (D) Its middle

6. Where did the research team work?

- (A) Toho University in Japan
- (B) Penn State in the U.S.
- (C) A lab in Brazil
- (D) A zoo in Australia

Answer key (try the quiz first!): 1: D 2: B 3: C 4: C 5: C 6: A

Questions to think about

MY THOUGHTS (at least 20 words):